

Assignment 2 Case Study

Stage 2 Tutorial From Problems to Requirements

Week 5 | 60 minutes

Learning goals

- Analyse stakeholders.
- Distinguish functional and non-functional requirements.
- Recognise ambiguity and unsupported requirements.
- Define scope.
- Develop user stories and acceptance criteria.
- Critique AI-generated requirements.

Activity 1 - Stakeholder Map

Stakeholder	Need	Potential conflict
Patient	Book appointments and have accurate patient information and appointment details	Convenience may be hindered as a result of having to verify details or make cancellations
Healthcare Practitioner	Patient information and appointment details, clear schedule where patient appointments do not overlap	Practitioner availability may conflict with receptionists' requests to allocate a patient appointment with them
Receptionist	Record and patient information, create and cancel appointments without creating data redundancy	Validation rules that prevents duplicate and/or invalid bookings and patient information may slow down efficiency
Clinic Manager	A simple software system that manages clinic operations and ensure patient, staff and appointment information are correct	Clinic manager wants to maintain simplicity for their software whereas receptionists or practitioners may want more features
System Administrator	A simple software system than can be maintained, tested and conserve consistent data	To ensure data is consistent and accurate, adding more validations can affect receptionists' efficiency to manage patient and staff information

Activity 2 - Functional or Non-Functional?

Scenario	Functional or Non-Functional?
The system shall allow staff to cancel an appointment.	Functional
The system should remain responsive for the course-scale dataset.	Non-Functional
The system shall retain cancelled appointments.	Functional
Core business logic should be independently testable.	Non-Functional
The system shall search for a patient by ID.	Functional

Activity 3 - Repair Ambiguous Requirements

The system should be easy to use.

Problem: “Easy to use” is subjective and not measurable.

Clarification question: Which core tasks must staff be able to perform successfully? In addition, what metrics can be used to define easy?

Patient search should be fast.

Problem: “Fast” is a vague term that does not specify a measurable time limit.

Clarification question: What is the maximum acceptable response time for a patient search? Will the acceptable response time differ depending on how much patient and staff information is in the system?

The system should securely manage data.

Problem: “Securely manage” is broad and lacks technical constraints. To secure data may require complex security systems and/or multiple features to ensure that data is secure. This conflicts with the requirements of a simple software system.

Clarification question: What specific security features must be implemented into the software system?

Appointments should normally be easy to cancel.

Problem: “normally easy to cancel” is an ambiguous and subjective phrase that does not define what an atypical scenario could be.

Clarification question: What is classified as a normal cancellation? What specific steps are required to resolve cancellations?

Activity 4 - AI Requirements Audit

Classify each suggestion: Confirmed / Assumption requiring validation / Unsupported / Out of scope.

AI suggestion	Classification	Evidence / reason
Patients receive SMS reminders.	Unsupported	Not a requirement the client needs for their system however could support patient management and ensure patients arrive for their appointment
Facial recognition login.	Out of scope	The clinic does not state issues with log in systems for either staff or possibly patients. Adds complexity to the system
Receptionists create appointments.	Confirmed	Confirmed: A core feature needed for the clinic to manage their system.
Online payment.	Out of scope	Does not align with the core requirement which is that the clinic wants a system that manages patient and staff information.
Practitioners view schedules.	Assumption requiring validation	Requires practitioner's insights on the idea, however could possibly help address the problem with the limited visibility of practitioner availability
AI recommends treatments.	Out of scope	AI can not guarantee accurate treatments, and causes risks in the reliability of the clinic. In addition, the implementation of AI is not a requirement for the clinic.
Cancelled appointments remain in history.	Assumption requiring validation	The client identifies issues in cancellations and reliable appointment history, but does not specifically clarify whether they want to retain appointment details if they have been cancelled

Exit question

Why is 'AI suggested it' not sufficient evidence for a requirement?

“AI suggested it” is not sufficient evidence to determine a requirement as it does not consider specific stakeholder needs in which AI tends to create out-of-scope suggestions that extend beyond the client’s requirements, limitations and budget.